Anirban Ray

Final-year PhD Student in Computer Science

Computational Image Restoration | Generative AI for Microscopy

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Profile

Final-year Computer Science PhD student developing generative AI and deep learning methods for computational microscopy. Research focus: inverse problems, image restoration, dehazing, super-resolution, denoising, uncertainty estimation, and posterior sampling. Prior industry experience as an AI researcher at Hitachi R&D in Tokyo.

Education

PhD in Computer Science — **2022–2026 (expected)**

Technische Universität Dresden & Human Technopole — *Dresden, Germany / Milan, Italy*

- Advisor: Dr. Florian Jug; Thesis Advisory Committee: Prof. Dr. Ivo F. Sbalzarini and Dr. Virginie Uhlmann.
- Thesis: *Generative Image Restoration for Microscopy.*
- Selected outputs: ResMatching (IEEE ISBI 2026 oral), HazeMatching (CVPR 2026 Findings), and RESOLFT deep-learning imaging (under review).

Master of Engineering in Computer Science — **2016–2018**

Nagoya Institute of Technology — *Nagoya, Japan*

- Advisors: Prof. Jun Sato and Prof. Fumihiko Sakaue.
- Thesis: *Modeling the Feature Evolution in CNNs using LSTM*; focus on computer vision and deep learning.

Research Student — **Oct. 2015–Mar. 2016**

Nagoya Institute of Technology — *Nagoya, Japan*

Bachelor of Technology in Computer Science and Engineering — **2011–2015**

Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology — *Chennai, India*

Research Experience

RESOLFT time-lapse imaging empowered by deep learning — **2024–Present**

Human Technopole; collaboration with the Testa Lab — *Milan, Italy*

- Applied deep learning restoration to low-SNR, sub-sampled RESOLFT nanoscopy, enabling 5× longer imaging with 10× lower light dose or 4× faster imaging while preserving approximately 60 nm resolution.
- Output: Research Square preprint, under review.

ResMatching: Noise-Resilient Computational Super-Resolution via Guided Conditional Flow Matching — **2024–2025**

Human Technopole — *Milan, Italy*

- Developed guided conditional flow matching for microscopy super-resolution, unifying denoising, uncertainty estimation, and posterior sampling.
- IEEE ISBI 2026 oral presentation; GitHub, arXiv, project page.

HazeMatching: Conditional Flow Matching for Microscopy Dehazing — **2023–2024**

Human Technopole — *Milan, Italy*

- Created a conditional flow-matching framework for microscopy dehazing, widefield-to-confocal restoration, and Expansion Microscopy super-resolution.
- CVPR 2026 Findings; CV4Science Workshop poster; GitHub, arXiv.

Deep Learning for Microscopy Image Analysis **2018–2021**
Hitachi Ltd., Center for Technology Innovation *Tokyo, Japan*

- Built deep learning systems for biomedical microscopy and SEM image analysis, including segmentation, counting, classification, quantification, CISS publications, and granted patents.

Modeling the Feature Evolution in CNNs using LSTM **2016–2018**
Nagoya Institute of Technology *Nagoya, Japan*

- Modeled feature evolution across CNN layers using LSTM networks for image classification.

Professional Experience

Associate Researcher **Apr. 2018–Jan. 2022**
Hitachi Ltd., R&D Group, Center for Technology Innovation *Tokyo, Japan*

- Delivered deep learning and computer vision systems for industrial R&D and biomedical microscopy image analysis.

Engineering Intern, AR Applications **Sept. 2016**
Sun Corporation *Konan, Japan*

Engineering Intern, Cloud Services **Aug. 2014–Jan. 2015**
Machine Pulse (Mahindra Teqo) *Mumbai, India*

Publications and Preprints

[1] **Anirban Ray et al.**. ResMatching: Noise-Resilient Computational Super-Resolution via Guided Conditional Flow Matching. *IEEE ISBI 2026*. Accepted; oral presentation.

[2] **Anirban Ray et al.**. HazeMatching: Conditional Flow Matching for Microscopy Dehazing. *CVPR 2026 (Findings)*. Accepted; CV4Science Workshop poster.

[3] **Minet, Ray et al.**. RESOLFT time lapse imaging empowered by deep learning. *Research Square preprint*. Under review.

[4] **Kakishita, Ray et al.**. Quantitative Analysis System for Bacterial Cells in SEM Image using Deep Learning. *CISS 2021*. pp. 1–6.

[5] **Kakishita, Ray et al.**. Deep Learning Based Bacteria Classification from SEM Images Using a Combination of Membrane and Internal Features. *CISS 2022*. pp. 13–18.

Patents

- Inventor on 3 granted Hitachi High-Tech image-processing patents: US12327363B2, US12211213B2, and EP3961562B1.

Presentations and Posters

- **Invited talk**: Scientific Inverse Problems in the Age of Generative AI. AI & Advanced Computing Institute, Schmidt Sciences, New York, USA, Jun. 2026.
- **Talk and poster**: ResMatching. IEEE ISBI 2026, London, UK, Apr. 2026.
- **Poster**: HazeMatching. IEEE/CVF CVPR 2026 and CV4Science Workshop, Denver, CO, USA, Jun. 2026.
- **Invited talk**: Introduction to Diffusion Model for Microscopy. EMBO-DL4MIA, Milan, Italy, May 2024.

Teaching and Mentoring

Teaching Assistant, Deep Learning for Microscopy Image Analysis (DL4MIA) **2022, 2023, 2024**
Human Technopole *Milan, Italy*

- Mentored participants, supported coding sessions and projects, and delivered an invited lecture in the 2024 edition.

Teaching Assistant, Deep Learning at MBL (DL@MBL) **2023, 2024**
Marine Biological Laboratory *Woods Hole, USA*

- Led hands-on support for deep learning in biological imaging; co-mentored project work linked to NeurIPS 2024 and RESOLFT microscopy outputs.

Professional Service

- Reviewer, AI4Life Open Call, 2024.

Awards, Honors, and Selected Training

- **Selected Participant, Optical Microscopy & Imaging in the Biomedical Sciences (OMIBS)**, Marine Biological Laboratory, USA, 2023; team won 3rd place in microscopy imaging.
- **Selected Participant, International Computer Vision Summer School (ICVSS)**, Italy, 2017; selected among 150 global students.
- **Aichi Monozukuri Scholarship**, Aichi Prefectural Government, Japan, 2015; full funding for master's studies, selected among top 10 students in Asia.
- **Selected Participant, Undergraduate Summer School on Computer Science**, Indian Institute of Science, Bengaluru, India, 2013.

Projects and Software

- **ResMatching**: Open-source implementation, data, models, project page, and quickstart notebook for guided conditional flow matching in microscopy super-resolution.
- **HazeMatching**: Open-source implementation, data, models, and quickstart notebook for conditional flow matching in microscopy dehazing and Expansion Microscopy super-resolution.

Skills

Research	Computational microscopy; image restoration; inverse problems; generative AI; conditional flow matching; diffusion models; denoising; dehazing; super-resolution; uncertainty estimation; posterior sampling; segmentation; computer vision.
Programming	Python; PyTorch; OpenCV; NumPy; Bash; Linux; Git; SLURM; LaTeX.
Languages	Bengali (native); Hindi (fluent); English (fluent); Japanese (conversational); Tamil (elementary); Italian (elementary).